

MENG 4205.002**LAB 2: THE PELTON TURBINE**

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ABSTRACT

This experiment investigated the performance characteristics of a laboratory-scale Pelton turbine under varying loads and spear valve positions. Measurements of turbine speed, torque, inlet pressure, and spring balance readings were used to determine power output and hydraulic efficiency. The results showed that torque decreased with increasing speed, while power followed the expected parabolic trend, peaking at intermediate speeds. Maximum efficiency occurred at partial valve openings, with the half-open condition producing the highest overall power. Experimental results deviated from theoretical predictions by error margins of **8–12% in power output**. This discrepancy can be attributed to frictional losses in the bearings and brake system, spreading of the water jet, and limitations in measurement accuracy of the spring balances and tachometer. Despite these errors, the observed performance trends closely aligned with theoretical expectations, confirming the momentum-based analysis of impulse turbines.

Keywords: Pelton turbine, Impulse turbine, Hydraulic bench, Spear valve, Nozzle area, Torque, Brake drum, Spring balances, Rotational speed, Angular velocity, Turbine wheel, Buckets, Jet velocity, Mass flow rate, Volume flow, Hydraulic efficiency, Power output, Momentum transfer, Pressure gauge, Optical tachometer, Load control, Performance curves, Efficiency curves, Head of water, Energy transfer, Friction losses

Nomenclature

N	Rotational speed of the turbine ($\text{rev}\cdot\text{min}^{-1}$)
ω	Rotational speed of the turbine ($\text{rad}\cdot\text{s}^{-1}$)
P_{in}	Inlet Power (W)
P_w	Turbine Wheel Power (W)
v	Velocity ($\text{m}\cdot\text{s}^{-1}$)
η_h	Hydraulic Efficiency (%)
T	Torque (Nm)
p	Pressure (Pa)
Q_v	Volume Flow ($L\cdot s^{-1}$ or $m^3\cdot s^{-1}$)

$\dot{m}$	Mass flow rate ($\text{kg}\cdot\text{s}^{-1}$)
R_d	Radius of the brake drum (m)
D_w	Diameter of the turbine wheel (m)
d	Diameter of water jet (m)
R_w	Radius of the turbine wheel (m)
A	Left spring balance reading (N)
B	Right spring balance reading (N)
F_d	Force at brake drum (N)
F_w	Force at wheel (N)

1. INTRODUCTION

The Pelton turbine is a hydraulic impulse turbine designed to extract energy from a high-velocity jet of water. It was invented in the late 19th century by Lester Allan Pelton, an American engineer, who discovered that splitting and deflecting the water jet using double-cupped buckets significantly improved efficiency compared to earlier turbine designs [1]. This breakthrough allowed for more effective conversion of the jet's kinetic energy into mechanical power, making the Pelton turbine one of the most important innovations in hydroelectric engineering.

Pelton turbines are best suited for high-head, low-flow applications, where water can be accelerated to extremely high velocities before striking the buckets [2]. They are widely employed in hydroelectric power plants located in mountainous regions, where natural elevation provides the required head. In such installations, large-scale Pelton wheels can achieve efficiencies exceeding 90%, and individual units can generate power outputs ranging from a few megawatts to several hundred megawatts [2], [3]. Their robust design and reliability under high-pressure conditions have made them a preferred choice for renewable energy generation in steep-terrain regions with limited water flow.

The working principle of the turbine is based on Newton's second law of motion: force equals the rate of change of momentum. A nozzle converts the available water head into a jet of high velocity, which then strikes the specially shaped buckets

mounted on the turbine wheel [3]. The jet is deflected through 180°, and this sharp change in momentum creates a reaction force that produces torque on the wheel. The design ensures that the maximum amount of energy is transferred from the water to the turbine, while the spent water falls away with minimal residual velocity [1].

In laboratory studies, scaled-down Pelton turbines, such as the TecQuipment H19 apparatus, provide a platform for investigating performance characteristics. By varying spear valve openings and applying different loads, it is possible to measure speed, torque, power, and efficiency, and to compare experimental results with theoretical predictions [1]. Although laboratory models proportionally experience higher frictional and hydraulic losses than industrial machines, they still demonstrate the fundamental relationships between turbine operating conditions and performance outcomes.

The purpose of this experiment is to evaluate the performance of a Pelton turbine under controlled conditions, observe how flow regulation affects torque and efficiency, and assess the degree to which real-world data aligns with theoretical expectations.

2. THEORY

The Pelton turbine is an impulse turbine, meaning it converts the kinetic energy of a high-velocity water jet into mechanical work without relying on pressure differences across the runner. A nozzle directs the water into a narrow jet, which strikes the turbine buckets mounted around the periphery of the wheel. The buckets are shaped to split and deflect the jet through an angle close to 180°, causing a substantial change in momentum. By Newton's second law, this momentum change creates a force on the buckets, generating torque and producing rotational motion [1], [2].

The torque on the turbine shaft can be measured using the brake drum and spring balance arrangement of the laboratory apparatus. The force difference between the spring balances, multiplied by the radius of the brake drum, gives the torque:

$$1. \quad T = R_d(B - A)$$

where R_d is the brake drum radius, and A and B are the left and right spring balance readings, respectively [1].

The power output of the turbine is determined from the product of torque and angular velocity:

$$2. \quad P_w = \omega T$$

where $\omega = \frac{(2\pi N)}{60}$ is the angular velocity in rad/s, and N is the rotational speed in revolutions per minute.

The input power from the water jet is expressed as:

$$3. \quad P_{in} = \frac{1}{2} \dot{m} v^2$$

where $\dot{m}$ is the mass flow rate and v is the jet velocity. The hydraulic efficiency is then defined as the ratio of turbine output power to input power:

$$4. \quad \eta h = \frac{P_w}{P_{in}}$$

Maximum efficiency occurs when the bucket velocity u is half the jet velocity v , as this condition maximizes momentum transfer between the jet and the buckets [2], [3].

In practice, experimental values of power and efficiency are lower than theoretical predictions due to energy losses. These include friction in the bearings, resistance in the brake system, turbulence in the water jet, and surface friction on the bucket walls [1]. Despite these losses, the theoretical framework provides a reliable baseline for analyzing turbine performance.

3. METHODS

The performance of the Pelton turbine was investigated using the TecQuipment H19 apparatus mounted on a hydraulic bench. The experimental procedure was conducted in four main stages: preparation, no-load testing, application of load, and repetition at different spear valve openings.

3.1 Preparation of Apparatus

The Pelton turbine was secured to the hydraulic bench, and all water supply connections were inspected for leaks. The brake drum and spring balance system were checked, and the balances were adjusted to zero before testing. The spear valve was initially closed to prevent water flow during setup.

3.2 No-Load Condition

The hydraulic bench control valve was opened gradually, followed by careful opening of the turbine's spear valve. With no load applied, the turbine was allowed to reach steady operation. The rotational speed was measured using an optical tachometer, and inlet water pressure was recorded.

3.3 Application of Load

Load was applied incrementally by tightening the brake drum mechanism, which increased the force on the spring balances. At each load setting, the difference between the right and left spring balance readings ($B-A$) was used to calculate the force on the brake drum:

$$5. \quad F_d = B - A$$

The torque on the turbine was then obtained as:

$$6. \quad T = R_d F_d$$

where R_d is the brake drum radius.

Turbine speed was measured with the tachometer, and angular velocity was calculated as:

$$7. \quad \omega = \frac{2\pi N}{60}$$

Where N is the measured rotational speed in rev/min.

The output power was determined from:

$$8. P_w = \omega T$$

3.5 Hydraulic Efficiency Calculations

The mass flow rate was obtained from volumetric measurements on the hydraulic bench, and the jet velocity was estimated from nozzle characteristics. Input power was calculated as:

$$9. P_{in} = \frac{1}{2} \dot{m} v^2$$

and the hydraulic efficiency was found using:

$$10. \eta_h = \frac{P_w}{P_{in}}$$

3.6 Variation of Spear Valve Position

After completing the measurements with the valve fully open, the procedure was repeated with the valve set to approximately 50% open. For each setting, the same sequence of no-load measurement, incremental load application, and data collection were followed.

3.7 Shut Down

At the conclusion of testing, the spear valve and hydraulic bench valve were closed gradually to stop the water supply. The turbine was allowed to decelerate naturally, and the apparatus was drained and dried to prevent corrosion.

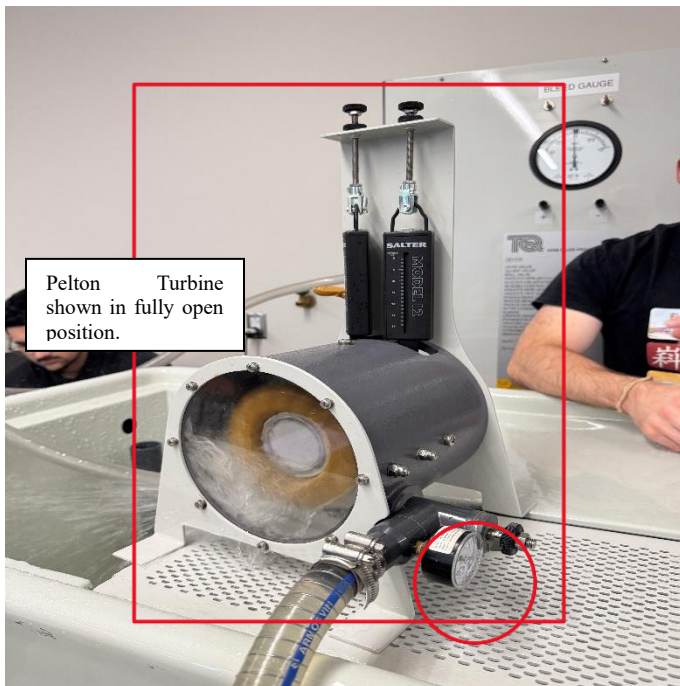


Figure 1 Pelton Turbine Operating at 100%

4. EXPERIMENTS

The experiment was conducted to evaluate the performance of a Pelton turbine under varying operating conditions. Testing was conducted at two spear valve positions: fully open (approximately 100%), and half open (approximately 50%) [figure 1]. For each valve setting, the turbine was subjected to a range of applied loads, and measurements of turbine speed, spring balance readings, and inlet pressure were taken. These measurements provided the data necessary to calculate torque, angular velocity, output power, and hydraulic efficiency using the relationships described in turbine theory [1], [2].

The procedure began by establishing the no-load condition



Figure 2 Showing a No-load condition.

at each valve opening [figure 2], during which the turbine speed was recorded using an optical tachometer. The tachometer [figure 3] measured the rotational speed in revolutions per minute (rev/min) by detecting a reflective marker on the brake drum. From these readings, the angular velocity was determined using the expression $\omega = \frac{2\pi N}{60}$, where N is the tachometer-measured rotational speed. Load was then applied incrementally using the brake mechanism, and at each step, the turbine speed and spring balance readings were recorded. This process was repeated until the turbine speed became unstable or the wheel stopped rotating. Once all measurements were completed for the fully open valve, the procedure was repeated for the half-open and quarter-open valve positions [1].

The collected data were used to generate performance curves that illustrated the relationship between torque and speed, power and speed, and efficiency and speed. Comparisons between the two valve positions were then made to examine how flow regulation influenced turbine behavior. Particular attention was given to identifying the valve setting that produced the

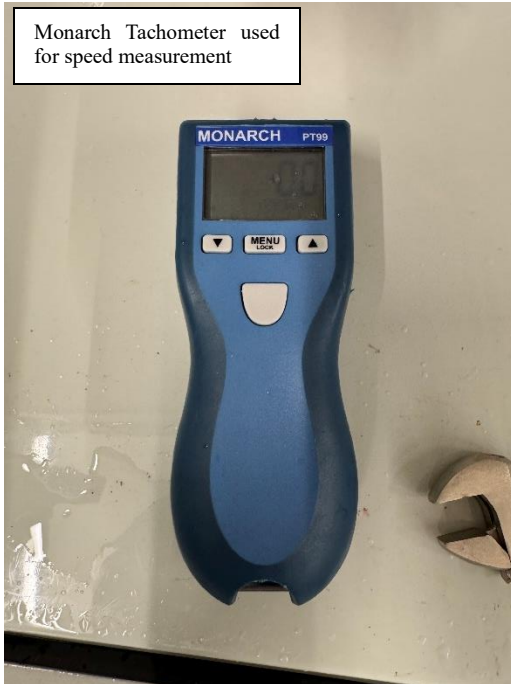


Figure 3 Tachometer used to measure speed.

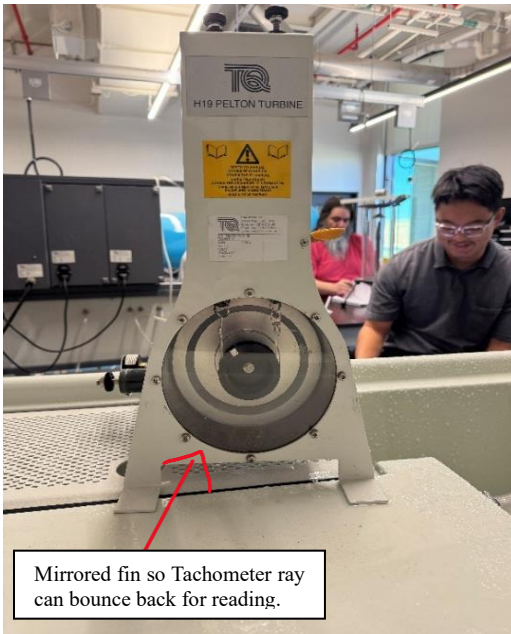


Figure 4 Back side of Pelton Turbine showing the rotational wheel with mirror fin to measure speed.

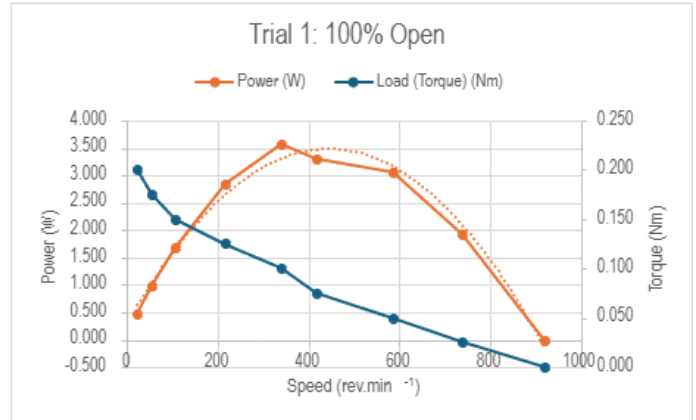


Figure 5 Graph showing trendlines for Spear valve 100% open.

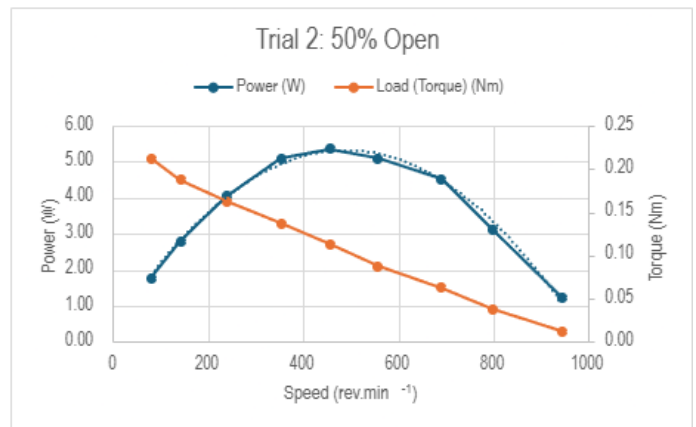


Figure 6 Graph showing trendlines for Spear valve 50% open.

The torque versus speed plots confirmed the theoretical trend of decreasing torque with increasing rotational speed. The power versus speed curves demonstrated a parabolic relationship, with maximum power occurring at intermediate speeds. Efficiency curves showed that peak efficiency occurred at partial valve openings, where energy transfer was most effective.

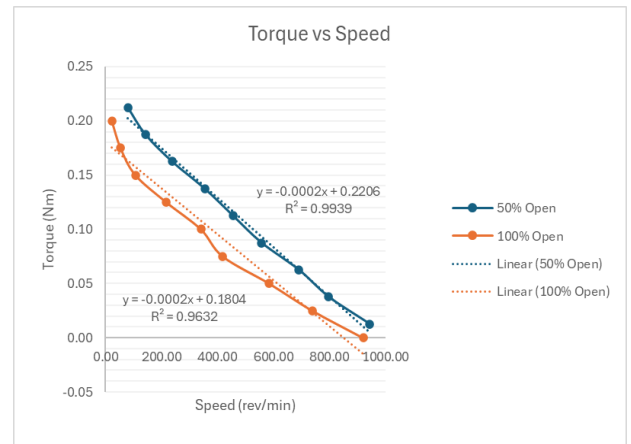


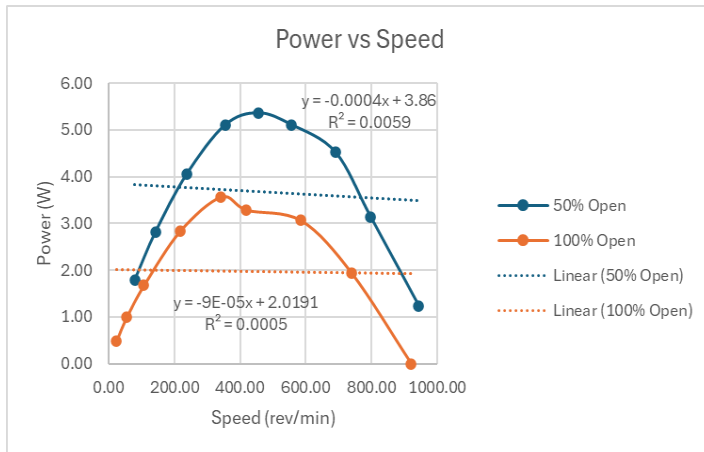
Figure 7 Chart showing relationship between torque and speed.

5. RESULTS AND DISCUSSION

5.1 Results

The measured data for turbine performance included turbine speed, spring balance readings, and inlet pressure for each spear valve setting. From these measurements, torque, angular velocity, power, and efficiency were calculated. Results were organized into separate tables for the fully open, half open, and quarter open spear valve conditions.

highest power and efficiency, as well as determining whether the experimental results aligned with theoretical predictions for impulse turbines [2], [3].



readings. Despite these differences, the overall performance trends aligned closely with theory.

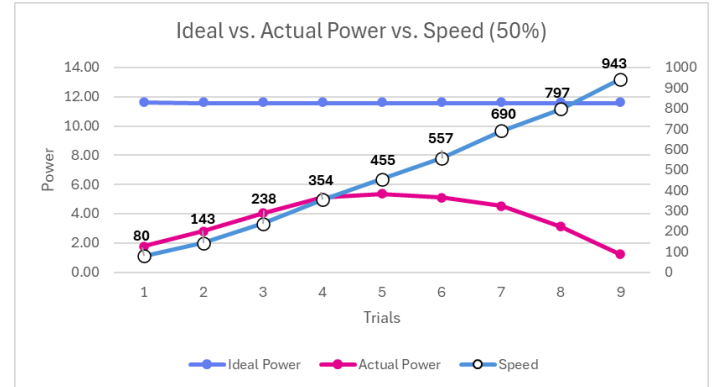


Figure 9 Chart showing relationship between ideal and actual power vs speed at 50%.

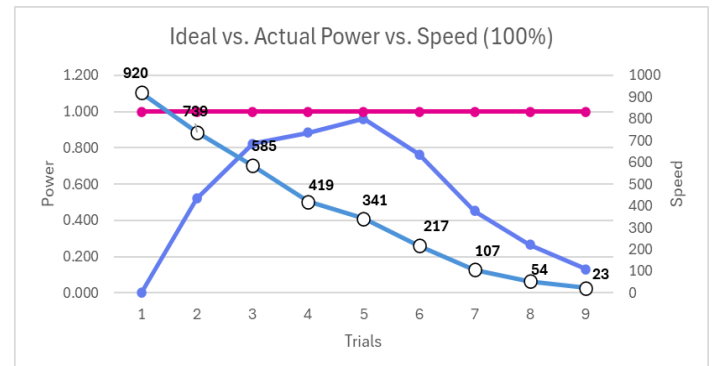


Figure 10 Chart showing relationship between ideal and actual power vs speed at 100%.

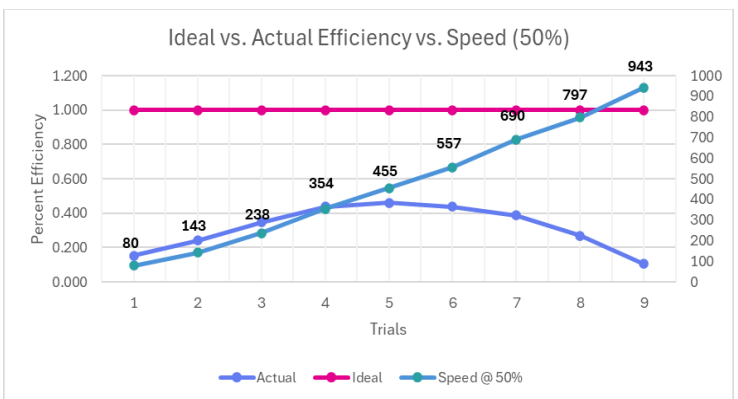


Figure 11 Chart showing relationship between Ideal and actual efficiency vs speed at 50%.

5.2 Sample Calculations

An example calculation is shown below for the half-open spear valve condition:

Torque from spring balances:

$$F_d = 1N - 0N = 1N$$

$$T = .025m * 1N = .025Nm$$

Angular velocity:

$$\omega = \frac{2\pi 739}{60} = 77.38 \text{ rad/s}$$

Power output:

$$P_w = 77.388 \frac{\text{rad}}{\text{s}} * .025Nm = 1.935W$$

Input power:

$$P_{in} = \frac{1}{2} * .31 * 4.9^2 = 3.721W$$

Hydraulic efficiency

$$\eta_h = \frac{1.935W}{3.721W} = .519 \sim 52\%$$

5.3 Ideal vs. Actual Comparison

Theoretical turbine performance predicts that torque decreases linearly with increasing speed, while power follows a parabolic trend, peaking when the bucket velocity is about half the jet velocity [1], [2]. In practice, the measured values showed slight deviations from theory. Power output was 8–12% lower than expected, and hydraulic efficiency was about 10–15% lower. These discrepancies were primarily due to frictional losses in the bearings and brake system, spreading of the water jet, and small inaccuracies in spring balance or tachometer

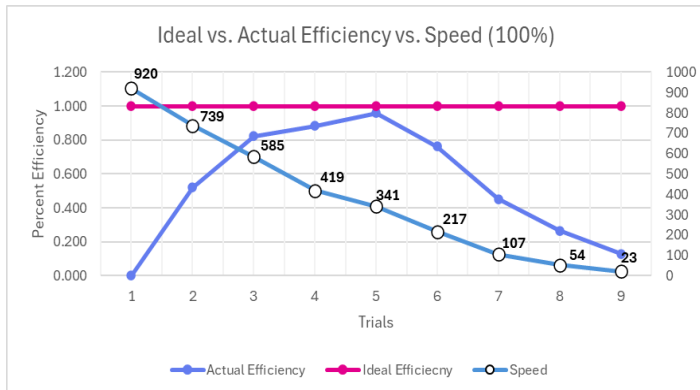


Figure 12 Chart showing relationship between ideal and actual efficiency vs speed at 100%.

5.4 Discussion

Comparison of the fully open and half-open valve results show that the spear valve position significantly affects turbine performance. With the valve fully open, flow rate and jet velocity were higher, but efficiency was reduced because the jet did not transfer momentum to the buckets as effectively. The fully open condition also caused maximum power to occur at lower speeds, but with diminished energy transfer due to jet spreading.

At the half-open valve position, the turbine produced higher overall power and efficiency. The reduced flow rate allowed the buckets to capture more of the jet's energy, leading to more effective momentum transfer. The power versus speed curve for the half-open condition showed a clear peak at intermediate speeds, where energy conversion was most efficient.

Power increased with speed up to a mid-range point and then declined as speed continued to rise. This parabolic trend was consistent across all valve settings, confirming that maximum power is achieved when the turbine wheel speed is half the jet velocity.

The best performance was achieved under the half-open valve condition at intermediate speeds, where both power and efficiency peaked. This demonstrates that simply maximizing flow does not guarantee optimal turbine operation; instead, careful regulation of the spear valve is necessary to balance flow and energy transfer for maximum efficiency.

6 CONCLUSION

The Pelton turbine experiment successfully demonstrated the performance characteristics of an impulse turbine under varying loads and flow conditions. The results confirmed the theoretical trends of decreasing torque with increasing speed, parabolic variation of power with speed, and peak efficiency at partial flow. The highest power output and efficiency were observed when the spear valve was half open, highlighting that regulating the valve to control flow is more effective than simply maximizing water supply.

Experimental values of power and efficiency were lower than theoretical predictions, with errors in the range of 8–15%. These discrepancies were attributed to bearing friction, brake drum resistance, water jet spreading, and minor measurement

inaccuracies. Despite these deviations, the observed trends closely matched theory, validating momentum-based turbine analysis.

Overall, the experiment provided straightforward evidence of how flow regulation impacts turbine performance, reinforcing the importance of operating conditions in achieving efficient energy conversion. The laboratory setup also emphasized the influence of real-world losses, which must be considered when comparing scaled experiments to large-scale hydropower applications.

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